



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SHREE VARNIKHA ENTERPRISES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

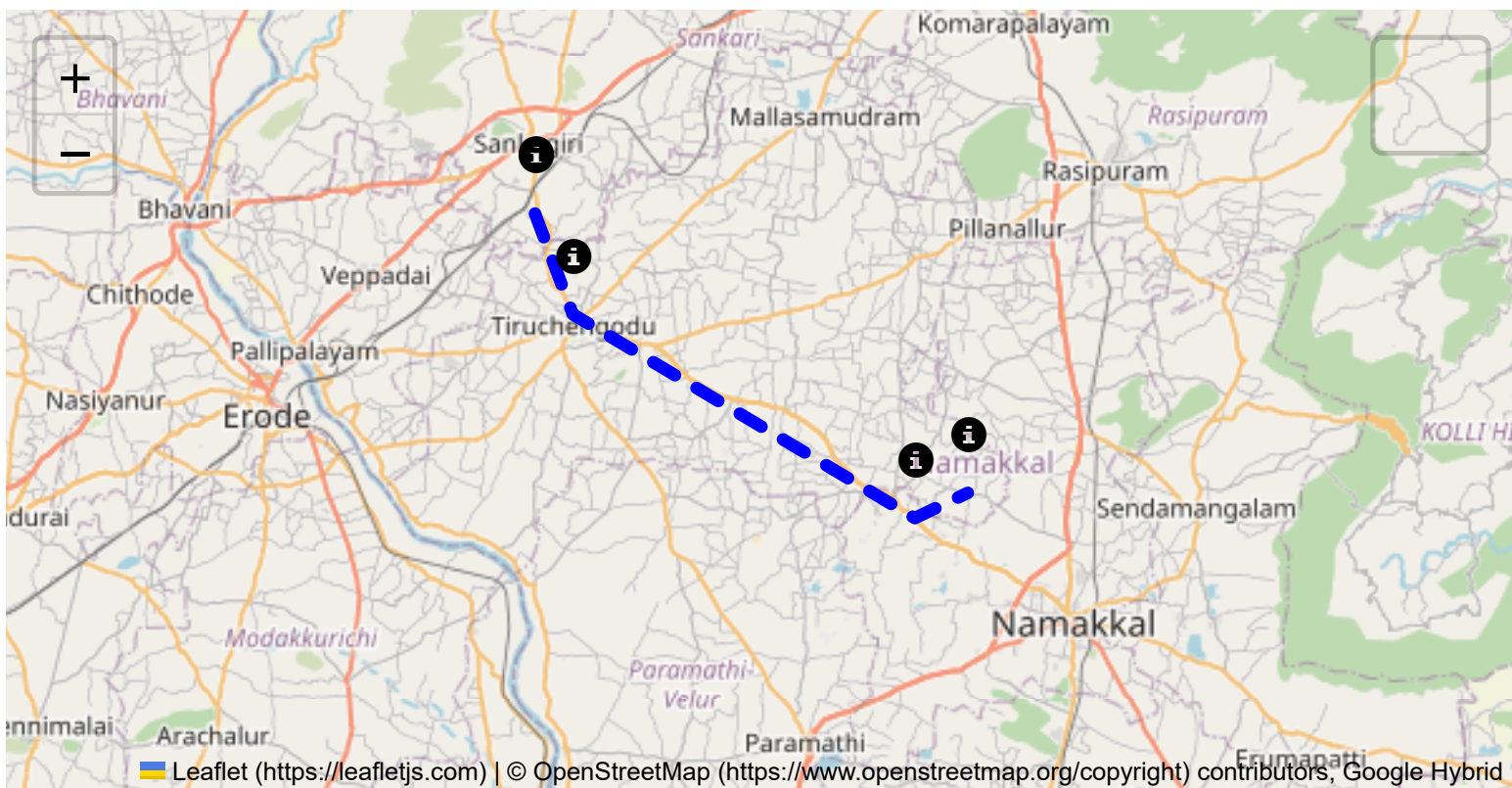
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

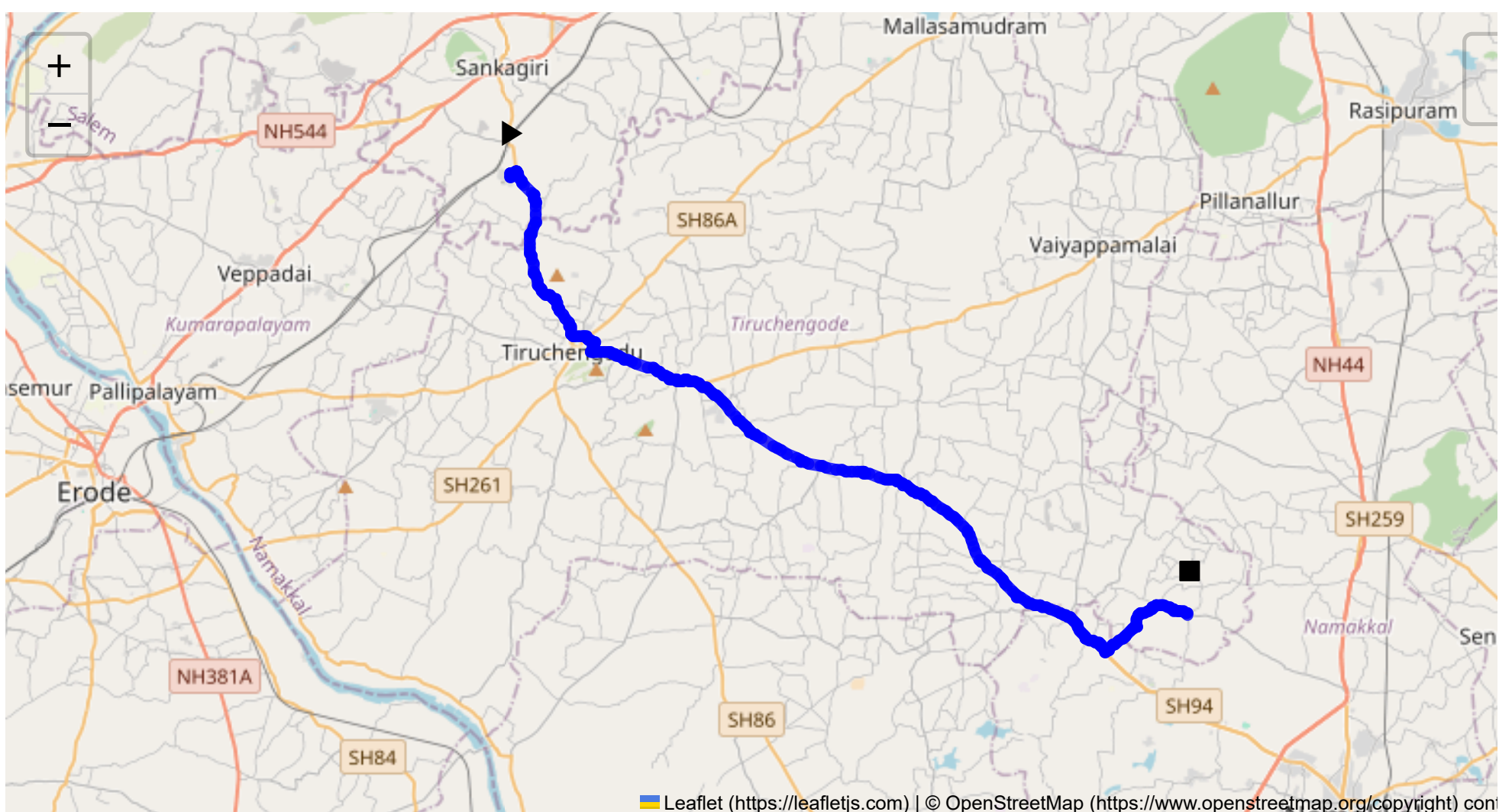
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 36.58 km
Estimated Duration: 0.8 hours
Adjusted Duration (Heavy Vehicle): 1.0 hours
Start: (11.4381, 77.8734)
End: (11.28831, 78.11018)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

- The route begins in Sangagiri, travels through Puthur and Tiruchengode, with intermediate stops including a Girls High School near Bharat Petrol Bunk and Namakkal, before reaching Sanagam Naickenpatti. It is primarily comprised of regional roads connecting small towns, with parts of the journey on state highways.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

- The region typically experiences a tropical climate. The main hazards include heavy monsoon rains (June to September) which can cause flooding, poor visibility, and slippery roads. During summer (March to May), extreme heat may lead to road surface degradation.

3. Analysis of Traffic Patterns:

- Traffic can be moderate to heavy during peak hours (8-10 AM and 5-8 PM), especially around Tiruchengode and Namakkal. Congestion is frequently observed near school zones and market areas, especially during school opening and closing times.

4. Assessment of Road Quality and Infrastructure:

- Generally, the state highways are in good condition, but regional roads may vary, with some parts potentially having potholes and narrow sections. Power outages can lead to non-functional traffic lights which can lead to temporary traffic hold-ups.

5. Suggestions for Alternative Routes for Emergencies:

- An alternative route could involve utilizing the NH47 (National Highway 47) which runs approximately parallel to the route. Although longer, it may offer better road conditions and faster travel times during emergencies.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Local regulations require heavy vehicles carrying hazardous materials to comply with specific timings, typically avoiding peak hours. Appropriate placarding and documentation are mandatory. Checkpoints may exist near populated areas to enforce safety standards.

7. Overview of Historical Incidents:

- While specific data may not be readily available, accidents involving heavy vehicles often occur due to poor road conditions and driver errors. Awareness and compliance with local driving norms are critical.

8. Environmental Considerations and Sensitive Areas:

- The route runs through some environmentally sensitive regions, including agricultural areas. Measures should be taken to prevent spills and contamination, especially near water bodies and farmland.

9. Analysis of Communication Coverage:

- Communication networks are moderately reliable across the route, but patches, especially in rural or more remote sections, could have weak or unstable mobile network coverage. Drivers should be aware of these potential dead zones.

10. Estimated Emergency Response Times:

- Emergency services in small towns often have varied response times, typically ranging from 20 to 40 minutes. Proximity to larger towns like Namakkal improves response availability.

11. Overall Summary of Risk Assessment:

- While traversing this route, risks mainly stem from inconsistent road conditions, weather influences, congestion, and communication issues. Careful planning, adherence to regulations,

and proactive communication with local authorities can mitigate these risks. With proper training and cautious driving, the route is manageable for transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.38346, 77.89975	15 KM/Hr
6	Turn	High	11.38190, 77.90254	15 KM/Hr
7	Turn	Medium	11.38182, 77.90259	30 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Blind Spot	Blind Spot	11.27561, 78.08152	10 KM/Hr
10	Turn	Medium	11.28062, 78.08817	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38346, 77.89975



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38182, 77.90259



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.27561, 78.08152



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.28062, 78.08817
